Lecture: Digital Humanities - Introductions
FS 2026

Contact persons in case of questions
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Course
Digital Humanities is a field of computer-based research in the humanities and social sciences. The lecture series provides an overview of the most important developments and research priorities in the Digital Humanities at Swiss universities. The interdisciplinary course is aimed at all students at the University of Bern.

The course will take place virtually and the individual lectures can be viewed online as videos. The videos are available on ILIAS and are organised in folders (course 1, course 2, …), there is no specific order in which you should work through the videos. There are two consultation session, one on April 21st (12.15-14.00) and one on May 27th (12.15-14.00). The counselling session will take place over Zoom; a link will be shared on ILIAS as well.

There is a MC exam taking place at the end of the course on June 1st, it will be provided over ILIAS. The exam will start at 0.05am and you'll have 24 hours to start it. As soon as you have started the exam, you will have 60mins to complete it, so do not start the exam later than 11pm on 1 June 2026.

Assessment
- Online MC exam, open book
- 20 questions randomly generated = individual exam for each person
- 4 possible answers, no negative points
- Scale: graded, 3 ECTS

Sample questions

Which of the following statements hold true, when following Walter Benjamins perspective? (Vera Chiquet)
- You can learn a lot by amusement and marvel about new media
- The will for education is the only way to gain knowledge about media
- Boredom dumbs down, curiosity (Kurzweile) enlightens
- Only by professional educational you can understand media processes

What form of machine learning is being used to train algorithms for Handwritten Text Recognition? (Tobias Hodel)
- Reinforcement Learning
- Always a combination of at least two of the mentioned technologies
- Supervised Machine Learning
- Unsupervised Machine Learning

Overview courses

Course	Name Lecturer	Institution	Topic	Further documents / papers
1	Andreas Fischer	HES-SO & Universität Fribourg	Text Recognition	-
2	Bianca Prietl	Universität Basel	Understanding Digitalization from Gender Perspectives	List at the end of the presentation
3	Cerstin Mahlow	ZHAW	Digital Linguistics	Aleksej V. Gladkij & Igor A. Mel'čuk (1973): Elemente der mathematischen Linguistik. Berlin: VEB Deutscher Verlag der Wissenschaften (Autorisierte Übersetzung aus dem Russischen, Original 1969, Moskau: Nauka). Linda S. Flower & John R. Hayes (1981): A Cognitive Process Theory of Writing. In College Composition and Communication, 32(4):365–387, doi: 10.2307/356600. John R. Hayes (2012): Modeling and Remodeling Writing. In Written Communication, 29(3):369–388, doi: 10.1177/0741088312451260. Daniel Perrin & MarcWildi (2009): Statistical modeling of writing processes. In C. Bazerman, R. Krut, K. Lunsford, S. McLeod, S. Null, P. Rogers, & A. Stansell (Eds.), Traditions of Writing Research, 378–393. Marlene Scardamalia & Carl Bereiter (1983): The development of evaluative, diagnostic and remedial capabilities in children's composing. In Margaret Martlew (Ed.) The psychology of written language. Developmental and educational perspectives, 67–95. Mike Sharples (1999): How We Write: Writing As Creative Design. Cerstin Mahlow & Michael Piotrowski (2022): Academic writing and publishing beyond documents. In: Proceedings of the 22nd ACM Symposium on Document Engineering (DocEng '22), 4pages. doi: 10.1145/3558100.3563840. Cerstin Mahlow & Robert Dale (2014). Production media: Writing as using tools in media convergent environments. In E. M. Jakobs & D. Perrin (Eds.), Handbook of writing and text production, handbooks of applied linguistics (Vol. 10, 209–230). Cerstin Mahlow (2015): A Definition of "Version" for Text Production Data and Natural Language Document Drafts. In: Proceedings of the 3rd International Workshop on (Document) Changes: Modeling, Detection, Storage and Visualization, 27-32. Py Kollberg & Kerstin Severinson Eklundh (2002): Studying writers' revising patterns with S-notation analysis. In Thierry Olive & C. Michael Levy (Eds.), Contemporary tools and techniques for studying writing, Studies in writing (Vol. 10, 89–104). Cerstin Mahlow, Małgorzata Anna Ułasik, & Don Tuggener (2022): Extraction of trans-forming sequences and sentence histories from writing process data: a first step towards linguistic modeling of writing. Reading and Writing, do: 10.1007/s11145-021-10234-6
4	Elena Chestnova	Università della Svizzera italiana	Digital Editing: Introduction	Examples of Digital Editions http://prhlt-kws.prhlt.upv.es/bentham/ Bentham Papers, UCL http://hallernet.org/ Albrecht von Haller https://www.darwinproject.ac.uk/ Darwin Correspondence Project

				https://fontane-nb.dariah.eu/index.html Theodore Fontane http://vangoghletters.org/vg/ Van Gogh Letters http://faustedition.net/ Goethe‘s „Faust“ https://www.menota.org/ Medieval Nordic Text Archive Resources and initiative related to editions TEI https://tei-c.org/ TEI Publisher https://teipublisher.com/ LEAF Writer https://leaf-writer.leaf-vre.org/ Sperberg-McQueen, C. M., and Claus Huitfeldt. ‘GODDAG: A Data Structure for Overlapping Hierarchies’. In *Digital Documents: Systems and Principles*, edited by Peter King and Ethan V. Munson, 139–60. Lecture Notes in Computer Science. Berlin, Heidelberg: Springer, 2004. Van Zundert, Joris, and Tara Andrews. “Apparatus vs Graph: New Models and Interfaces for Text.” In Interface Critique, edited by F. Hadler and J. Haupt, 183–205. Kaleidgramme 139. Berlin: Kulturverlag Kadmos, 2016. Kuczera, Andreas. “Digital Editions beyond XML - Graph-Based Digital Editions.” In Graph Technologies in the Humanities - Proceedings 2020, edited by Tara Andrews, Franziska Diehr, Thomas Efer, Andreas Kuczera, and Joris Van Zundert, 2020. Toschka, Patrick, Julian Jarosch, and Andreas Kuczera. “The Socinian Correspondence: A Graph-Based Digital Scholarly Edition.” In Graph Technologies in the Humanities - Proceedings 2020, edited by Tara L. Andrews, Franziska Diehr, Thomas Efer, Andreas Kuczera, and Joris Van Zundert, 15, 2020.
5	Gerhard Lauer	Universität Mainz	Digital Book Studies	-
6	Lukas Rosenthaler	Universität Basel	Open Linked Data & Semantic Web	-
7	Michael Piotrowski	Université de Lausanne	Theory and Epistemology of Digital Humanities	-
8	Noah Bubenhofer	Universität Zürich	Introduction into Corpus Linguistics	CLARIN Virtual Language Observatory: https://vlo.clarin.eu Corpora in Switzerland: https://clarin-ch.ch Noah Bubenhofer: Einführung in die Korpuslinguistik, https://www.bubenhofer.com/korpuslinguistik/ Swissdox@LiRI: Swiss media database for academic use, https://www.liri.uzh.ch/en/services/swissdox.html
9	Peter Fornaro	Universität Basel	Imaging Technologies in DH (German only, not relevant for exam)	-
10	Ramona Roller	ETH	Network Analysis	Newman, M. (2018). *Networks*. Oxford university press. Classic textbook on network analysis Historical Network Research. Community of researchers applying network analysis to historical data. They also have their own journal. Social Networks. Scientific journal with network analysis of data from the social sciences and sometimes the humanities. Other links are listed on the lecture slides.

11	Rico Sennrich	Universität Zürich & DARIAH	Natural Language Processing	Koehn, P. (2020).Neural Machine Translation. Cambridge University Press. Li, M., Lv, T., Cui, L., Lu, Y., Florêncio, D. A. F., Zhang, C., Li, Z., and Wei, F. (2021). Trocr: Transformer-based optical character recognition with pre-trained models. CoRR, abs/2109.10282. Ströbel, P. B., Volk, M., Clematide, S., Schwitter, R., Hodel, T., and Schoch, D. (2022). Evaluation of HTR models without ground truth material. In Proceedings of the Thirteenth Language Resources and Evaluation Conference, pages 4395–4404, Marseille, France. European Language Resources Association.
12	Selena Savić	FHNW	Digitization. Data Materialism in Digital Humanities	-
13	Stefan Münnich	Universität Basel	Digital Musicology	Link to slides with all links: https://tinyurl.com/digital-musicology-intro
14	Tobias Hodel	Universität Bern & DARIAH	Early Modern History after the Machine Learning Turn	Hodel, Tobias. „Chapter 6: Supervised and Unsupervised: Approaches to Machine Learning for Textual Entities“. In *Archives, Access and Artificial Intelligence. Working with Born-Digital and Digitized Archival Collections*, herausgegeben von Lise Jaillant. Digital Humanities Research 4. Bielefeld: Transcript, 2022. Hodel, Tobias. „Die Maschine und die Geschichtswissenschaft: Der Einfluss von deep learning auf eine Disziplin“. In *Digital History: Konzepte, Methoden und Kritiken Digitaler Geschichtswissenschaft*, herausgegeben von Karoline Dominika Döring, Stefan Haas, Mareike König, und Jörg Wettlaufer, 6:65–80. Studies in Digital History and Hermeneutics. Berlin, Boston: De Gruyter Oldenbourg, 2022. https://doi.org/doi:10.1515/9783110757101-004. Hodel, Tobias, David Schoch, Christa Schneider, und Jake Purcell. „General Models for Handwritten Text Recognition: Feasibility and State-of-the Art. German Kurrent as an Example“. *Journal of Open Humanities Data*, Journal of Open Humanities Data, 7 (2021). https://doi.org/10.5334/johd.46. Hodel, Tobias. „Konsequenzen der Handschriftenerkennung und des maschinellen Lernens für die Geschichtswissenschaft. Anwendung, Einordnung und Methodenkritik“. *Historische Zeitschrift* 316, Nr. 1 (2023): 151–80.
15	Tobias Wildi	FH Graubünden	Introduction to Archives and Digital Preservation	Digital Preservation Coalition: Digital Preservation Handbook. https://www.dpconline.org/handbook
16	Vera Chiquet	Universität Basel	Digital Photography	Fineman, Mia (2012) Faking It. Manipulated Photography before Photoshop. Metropolitan Museum of Art, New York. Siepmann, Eckhard (1983) Montage: John Heartfield, Elefanten Press. Chiquet, Vera (2018) Fake Fotos, transcript Bielefeld.
17	Yannick Rochat	Université de Lausanne	Game Studies	-